

# How to Use the CCS Data-Sufficiency Explorer

## A step-by-step guide to the interactive dashboard

**What this is.** An interactive page that answers one question for any of Georgia’s continuous count stations: *how many valid days does this station need in this calendar month before its monthly average can be published?*

**What you need.** A web browser. Nothing to install, no internet connection, no statistics background.

**Coverage.** All **233** stations, all **12** months — **2,796** station-months — under **six** different outage patterns.

This guide walks through the tool from the first double-click to reading the final number. Everything it describes is done by clicking. Nothing you do in the tool can change or damage the underlying data: the page only reads itself.

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## 1 The question the tool answers

A continuous count station reports traffic every day. When equipment fails, or a day is rejected in quality control, the month is left partly observed. The practical question follows immediately: *is what survived enough to publish a monthly average?*

Counting the surviving days does not settle it. Consider two Aprils that both kept exactly fourteen days:

- In the first, the fourteen are scattered across the month. Weekdays and weekends are both still represented, so the surviving days still describe the month.
- In the second, the counter died on the 7th and came back on the 24th. The fourteen survivors are two clumps at either end, and everything in the middle of the month is guesswork.

Same count, very different evidence. So the tool asks for two things — how many days survived, *and* how they went missing — and gives back a specific number of days for that combination.

**The number is guidance for a *future* month.** A month you already hold, and which came in complete, is simply reported: its average is known exactly and no prediction is involved. The recommendation tells you what to aim for next time.

## 2 Opening it

You will have been sent a folder. Unzip it anywhere you like — the desktop is fine — and inside you will find three things:

| File                                       | What it is                                    |
|--------------------------------------------|-----------------------------------------------|
| <b>GDoT Data-Sufficiency Explorer.html</b> | The tool. Double-click it.                    |
| dashboard_how_to_use.pdf                   | This guide.                                   |
| README.md                                  | A short plain-text summary of the same thing. |

**Double-click the first one.** It opens in your web browser and that is the whole procedure. There is nothing to install, no program to run first, no internet connection needed, and nothing you do leaves your machine.

- Everything the tool needs is inside that single file, including the data for all **233** stations. It is large for that reason, so the first load takes a few seconds. After that it is immediate.
- You can forward that one file to a colleague and it will work for them exactly as it does for you.
- On some computers `.html` files are set to open in a code editor rather than a browser, so you see markup instead of the dashboard. Right-click the file → **Open With** → your browser. You only need to do this once.

**If you have read Appendix E of the final report.** It refers to `gdot_dashboard.html` in a `dist` folder. That file has been renamed to **GDoT Data-Sufficiency Explorer.html** and moved to the top of this folder. Same tool, same data, same numbers — the instruction above is the current one.

### 3 Your first run, in five steps

This section produces a real answer. It uses station 153-0189 in April, which is the case study in the final report, so the numbers here can be checked against it.

#### Step 1 — Choose the station, the month, and the outage pattern

Everything starts in the control strip along the top.

| STATION                      | CALENDAR MONTH     | MISSINGNESS MECHANISM ⓘ |
|------------------------------|--------------------|-------------------------|
| 153-0189 — two way, US-129 ▼ | April — complete ▼ | Block-1 ▼               |

Set **Station** to 153-0189, **Calendar month** to **April**, and **Missingness mechanism** to **Block-1** (one unbroken outage). Leave  $\alpha$ ,  $\epsilon$  and  $R$  alone — Section 5 explains them, and the delivered values are the ones the report used.

Notice that the month is labelled **April — complete**. The list tells you what actually arrived, so you can see before choosing whether a month came in whole.

#### Step 2 — Read the six answer tiles

The row of tiles beneath the controls describes the station-month you just chose.

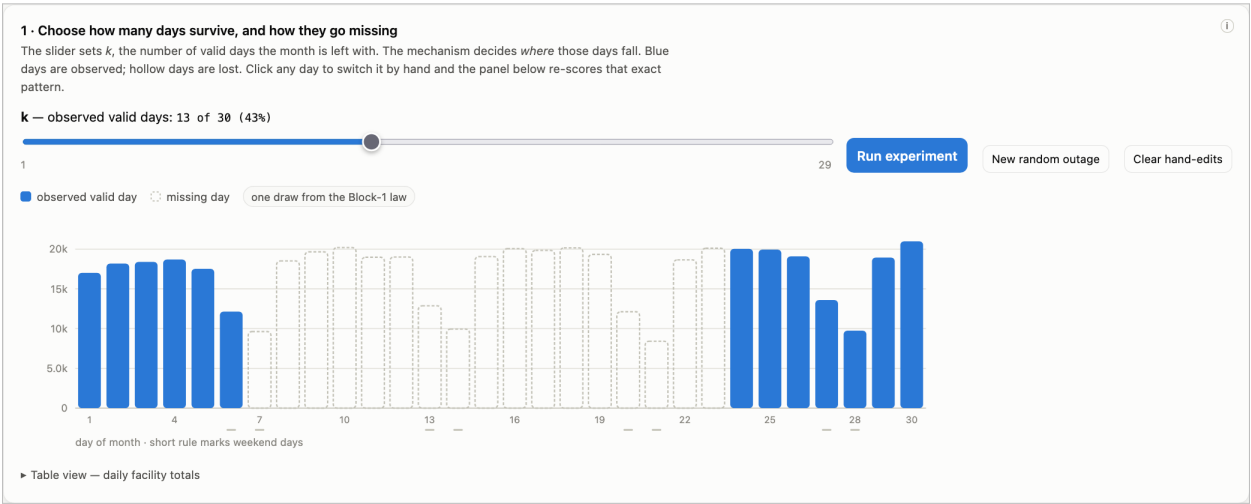
|                                                                                  |                                                                                                                          |                                                                               |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>FACILITY</b> ⓘ<br><b>two way</b><br>two way · FC 3 · urban · US-129           | <b>2024 TARGET STATE</b> ⓘ<br><b>Complete —<br/>reported exactly</b><br>30 of 30 strict-valid days                       | <b>EXACT MADT</b> ⓘ<br><b>17,025</b><br>vehicles/day, full calendar           |
| <b>BASELINE SCALE B</b> ⓘ<br><b>15,311</b><br>frozen from 2017–2019 design years | <b>RECOMMENDED PLANNING VALUE</b> ⓘ<br><b>13</b><br>max over 5 of 6 patterns · 43% of the month · 20 incl. weekend-heavy | <b>COHORTS (2022 / 2023)</b> ⓘ<br><b>41 / 43</b><br>cluster scope · preferred |

Read them left to right: this is a two-way station on US-129; April 2024 arrived **complete**, all 30 of 30 days valid; its true monthly average was **17,025** vehicles/day; its normal scale for April is **15,311**; the recommended planning value is **13** days; and **41 / 43** comparable stations supplied the evidence.

**The headline answer is the fifth tile: 13 days.** For this station, in April, thirteen valid days are enough to publish a monthly average that is both reliable and tight enough to act on — provided the days that went missing did so in one unbroken stretch.

Step 3 — See which days survive

The first tab, **Run an experiment**, shows the month itself.



The slider sets  $k$ , the number of valid days the month is left with; here it is at 13 of 30, which the tool also reports as 43% of the month. Solid blue bars are days that survived, and their height is that day’s traffic. Hollow dashed outlines are days that were lost. Because the pattern is Block-1, the lost days form one continuous run — the counter goes down on the 7th and comes back on the 24th.

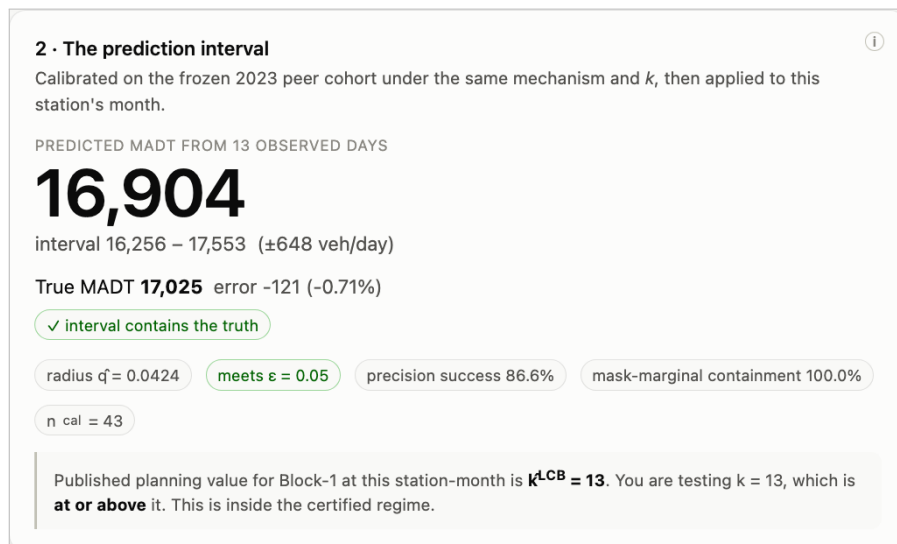
Three buttons sit beside the slider:

| Button                   | What it does                                                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Run experiment</b>    | Re-runs the simulation at the current settings.                                                                                     |
| <b>New random outage</b> | Draws a different outage of the same shape and the same length, so you can see how much the answer moves from one draw to the next. |
| <b>Clear hand-edits</b>  | Undoes any days you switched by hand and returns to a drawn pattern.                                                                |

You can also **click any individual day** to switch it on or off. That is how you test an outage you actually experienced, rather than a simulated one.

## Step 4 — Read the prediction interval

Directly below the calendar is the answer for the days now surviving.



From those 13 days the tool estimates the monthly average at **16,904** vehicles/day, and puts an interval of **16,256 – 17,553** around it — give or take 648 vehicles/day. Because this April was complete, the true answer is known: **17,025**. The estimate is off by 121 vehicles/day, or  $-0.71\%$ , and the interval contains the truth.

The row of small badges is the scorecard:

| Badge                     | Meaning                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------|
| radius $\hat{q}$          | How wide the interval is, as a fraction of the station's normal scale. Here 0.0424, i.e. about $\pm 4.2\%$ . |
| meets $\epsilon$          | Green when the interval came out tight enough against the target; red when it did not.                       |
| precision success         | Across all the simulated outages, how often the interval was that tight.                                     |
| mask-marginal containment | How often the interval contained the true value.                                                             |
| $n_{\text{cal}}$          | How many comparable stations the margin was learned from.                                                    |

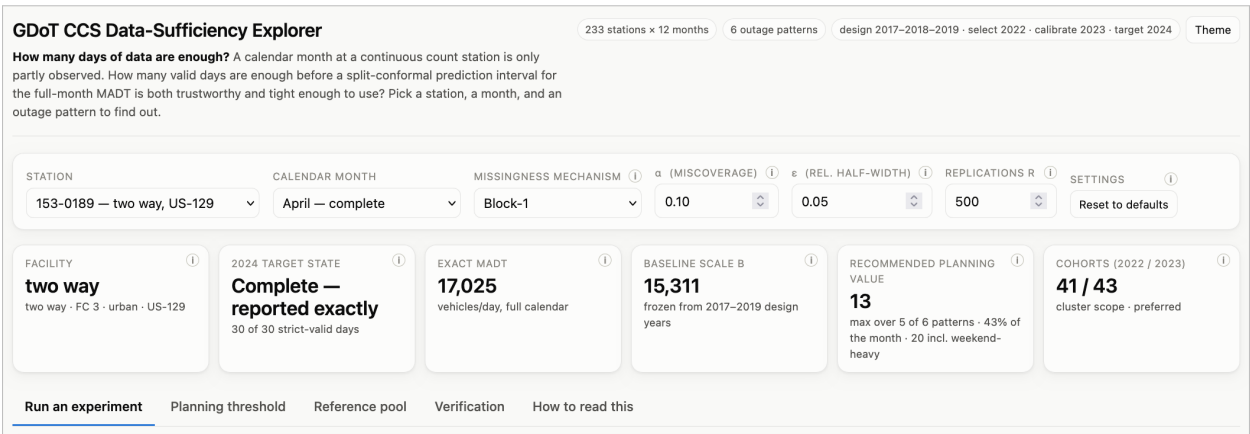
## Step 5 — Check you are inside the certified range

The grey note at the foot of that panel is the sentence that ties the experiment back to the recommendation:

*“Published planning value for Block-1 at this station-month is  $\hat{k}^{\text{LCB}} = 13$ . You are testing  $k = 13$ , which is **at or above** it. This is inside the certified regime.”*

That is the whole workflow in one line. Above the recommended number, the guarantee applies. Below it, it does not — and the tool says so plainly rather than leaving you to work it out. Section 10 shows what going below looks like.

## 4 The screen, region by region

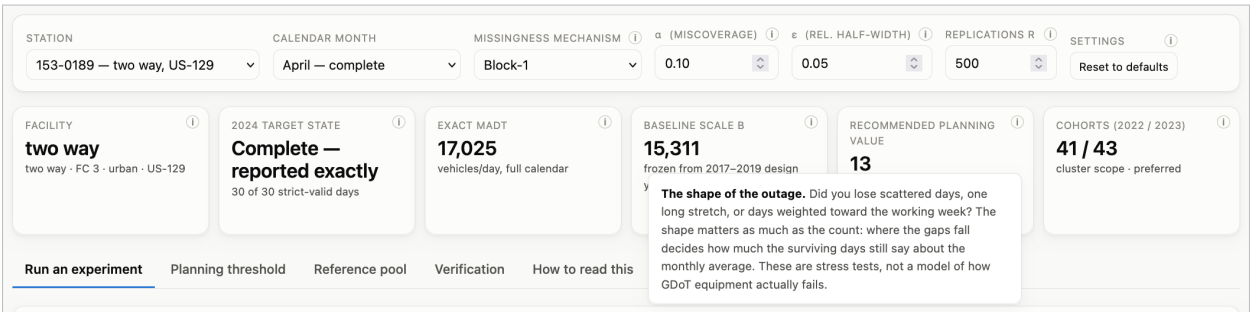


The whole tool fits on one screen. This view is for orientation — each region is shown full size in the sections that follow.

| Region        | What it is for                                                                                                                 |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Title bar     | The question, three badges recording the scope of the delivery and which years played which role, and the <b>Theme</b> button. |
| Control strip | Station, month, outage pattern, and three statistical settings.<br>Everything below reacts to these.                           |
| Answer tiles  | Six facts about the chosen station-month, including the recommended day count.                                                 |
| Tabs          | Five views of the same station-month, from a hands-on experiment to the evidence behind the published number.                  |
| Working area  | Whatever the selected tab shows.                                                                                               |

### The circled i markers

Every control, tile and panel heading carries a small circled **i**. Hover it — or reach it with the keyboard and pause — and it explains that item in plain language.



Nothing is hidden behind these. They repeat and expand what is already on screen, so the tool can be used without this guide open beside it.

## Light or dark

The tool follows whatever your computer is set to, so on a machine in dark mode it opens dark. Every screenshot in this guide is light. Nothing else differs — the same controls in the same places showing the same numbers — so if the colours on your screen do not match the pictures here, that is the reason, and nothing is wrong. The **Theme** button at the top right switches between the two whenever you like. It is not remembered: reopening the file returns to your computer’s own setting.

## The tables under the charts

Most panels carry a small grey **Table view** line beneath the picture. Click it and the numbers behind that chart open underneath. Some start open and some start closed, so if a figure quoted in this guide is not on your screen, that line is usually where it is.

## 5 The control strip

$\alpha$  (MISCOVERAGE) ⓘ

0.10

⬆️⬇️⬆️

$\epsilon$  (REL. HALF-WIDTH) ⓘ

0.05

⬆️⬇️⬆️

REPLICATIONS  $R$  ⓘ

500

⬆️⬇️⬆️

SETTINGS ⓘ

Reset to defaults

| Control                      | Ships as | What it does                                                                                                                                                                                                                                                              |
|------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Station                      | —        | Which counting station. All 233 that delivered data in 2024.                                                                                                                                                                                                              |
| Calendar month               | —        | Which month, labelled with what actually arrived.                                                                                                                                                                                                                         |
| Missingness mechanism        | Block-1  | The <i>shape</i> of the outage. The most important control, and the one Section 6 is about.                                                                                                                                                                               |
| $\alpha$ (miscoverage)       | 0.10     | How often the interval is allowed to be wrong. At 0.10 it is built to contain the true monthly average at least 90 times out of 100. Lower it for a stronger promise and the interval widens to keep it. <i>This dial sets reliability, not tightness.</i>                |
| $\epsilon$ (rel. half-width) | 0.05     | How tight the answer has to be. At 0.05 the interval may reach no more than 5% above or below the estimate — a 10% window. It is a share of the station’s own normal traffic, which is what lets a quiet county road and a busy interstate be judged on the same footing. |
| Replications $R$             | 500      | How many times the outage is simulated. Each replication knocks out a different set of days and works the answer out again. More replications give steadier numbers and take longer. <i>The report used 1,000; 500 is ample for exploring.</i>                            |
| Reset to defaults            | —        | Puts everything back. See below.                                                                                                                                                                                                                                          |

**Reliability and tightness are different things**, and this is the one idea worth pausing on. An interval can be perfectly reliable and uselessly wide (“the average is somewhere between 5,000 and 40,000” is almost certainly true and completely useless), or narrow and wrong.  $\alpha$  governs the first,  $\epsilon$  the second, and the recommended day count is the smallest number of days that satisfies *both*, often enough.

**What the dials move, and what they do not.** Changing  $\alpha$ ,  $\epsilon$  or the replication count *re-runs the experiment*: the interval, its width, and the success rates below all recompute from the underlying data.

The **recommended planning value** does not follow them. It was certified once, at a fixed set of values, and it keeps them:

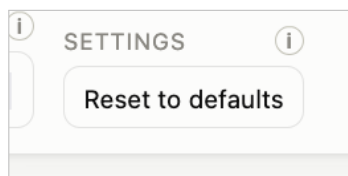
|                    |                   |                              |
|--------------------|-------------------|------------------------------|
| $\alpha = 0.10$    | $\epsilon = 0.05$ | success target = 90%         |
| 1,000 replications | seed 20260719     | six declared outage patterns |

Those are the values behind every number in the delivered spreadsheet and in the report. And it is worth being clear that the recommendation is not *independent* of them — it plainly is not. Loosen  $\alpha$  and fewer days would do; tighten  $\epsilon$  and you would need more.

The reason is cost. Deriving that number means re-running the entire selection experiment: every day count, every outage pattern, a thousand replications each, and then a simultaneous statistical correction across all of those cells at once. That is a far heavier calculation than anything the page does live. So the figure stays as it was certified, at the values in the table above — which is also what makes it match the report and the spreadsheet.

**If you do want to explore other settings.** Set  $\alpha$  and  $\epsilon$  where you want them, then move the day count and watch the *precision success* figure on the interval panel. That tells you how often a month with that many days would come out tight enough under *your* settings. It is a sound indication, but it is not a certified value: it comes from one fold and carries no correction for the many day counts you tried.

## Reset to defaults



Returns the outage pattern,  $\alpha$ ,  $\epsilon$ , the replication count, the day count and the random draw to the values the tool ships with, and forgets any days you switched by hand. **Your chosen station and month stay put**, so you can experiment freely and get back to a known state without losing your place.

## When you have changed something

Move  $\alpha$ ,  $\epsilon$  or the replication count away from the delivered values and the tool says so: an amber strip appears under the control row naming what you changed, and the **Planning threshold** tab is marked to warn you that the certified number on it was not computed at your settings. Neither is an error. They are there so a figure read off a changed screen is never mistaken for the delivered one. **Reset to defaults** clears both.

## 6 The six outage patterns

This is the least familiar control and the one that does the most work. It does not ask *how many* days went missing; it asks *which* ones. The six choices are drawn below. Every panel is the *same* April, and in every panel *exactly fourteen* days survive — only their arrangement changes.

■ observed day    ■ missing day

### MCAR

*Random dropouts*

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

Communication glitches drop isolated days, independently of each other.

### Block-1

*One outage*

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

A power failure keeps the counter down for one unbroken run of days.

### Block-2

*Two outages*

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

A modem fault, then a power-supply fault days later: two separate runs.

### Weekday-heavy

*Working week lost*

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

Weekday road-works: most of the loss lands Mon–Fri, and weekends are hit least.

### Weekend-heavy

*Weekends lost*

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

Routine weekend maintenance disconnects the sensor every Sat–Sun.

### Markov

*Bursty failures*

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

A loose connection: once it drops it tends to stay down, and the reverse.

- **MCAR** is the mildest case. The surviving days are spread over the whole month, so they still represent it fairly.

- **Weekend-heavy** is usually the hardest, for the reason set out in Section 11: a month holds only eight or nine weekend dates, and losing them removes a whole block of the weekly rhythm.
- The other four sit in between, and which of them bites hardest depends on the station’s own weekly pattern.

These are deliberate stress tests, not a model of how GDOT equipment actually fails. They exist so a recommendation can be tied to a *stated* assumption instead of an average over unknown failure modes.

**Which one should you pick?** If you are sizing a station for general planning, use the delivered recommendation, which already takes the worst of five of them. If you know the failure mode you are worried about — a modem that drops out intermittently, a planned closure — pick that pattern and read its number directly.

## 7 The six answer tiles

| Tile                                 | What it tells you                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Facility</b>                      | What is being counted. A two-way station needs both directions reporting; one direction alone is a different quantity and is never used.                                                                                                                                                              |
| <b>2024 target state</b>             | What actually arrived. <i>Complete</i> means every day passed quality control, so the average is simply reported. <i>Incomplete</i> means days are missing — the situation this tool is for. <i>No 2024 record</i> means nothing usable arrived for that month at all. Section 8 covers the last two. |
| <b>Exact MADT</b>                    | The true monthly average, shown only when the month came in complete, because that is the one case where the answer is known.                                                                                                                                                                         |
| <b>Baseline scale <math>B</math></b> | This station’s normal traffic for this month, fixed from its 2017–2019 history. Errors and interval widths are quoted as a percentage of it.                                                                                                                                                          |
| <b>Recommended planning value</b>    | The headline answer: how many valid days to aim for. The small print beneath it says what fraction of the month that is, and what the number would be if weekend-heavy were included — see Section 11.                                                                                                |
| <b>Cohorts (2022 / 2023)</b>         | How many comparable stations supplied the evidence, and from which grouping. Below <b>9</b> no usable interval exists at all; <b>19</b> or more is preferred for a stable width.                                                                                                                      |

## 8 When the month is not complete

The walkthrough used April at 153-0189, which arrived whole. Most months did, but not all, and the screen behaves differently for the rest. Across the 2,796 station-months in the tool:

| 2024 target state     | Months | What you see                                                                                                                                                                                 |
|-----------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Complete</b>       | 2,059  | The average is known, so it is reported exactly. You can still simulate an outage and watch the interval; because there is a true value, the tool can also tell you whether it was captured. |
| <b>Incomplete</b>     | 506    | The calendar shows the days that actually arrived rather than a simulated outage. There is no exact average and no hit-or-miss check, because there is no complete month to check against.   |
| <b>No 2024 record</b> | 231    | There is no target month to draw at all. The recommended day count is still shown, because it is guidance for a future month and does not depend on this one.                                |

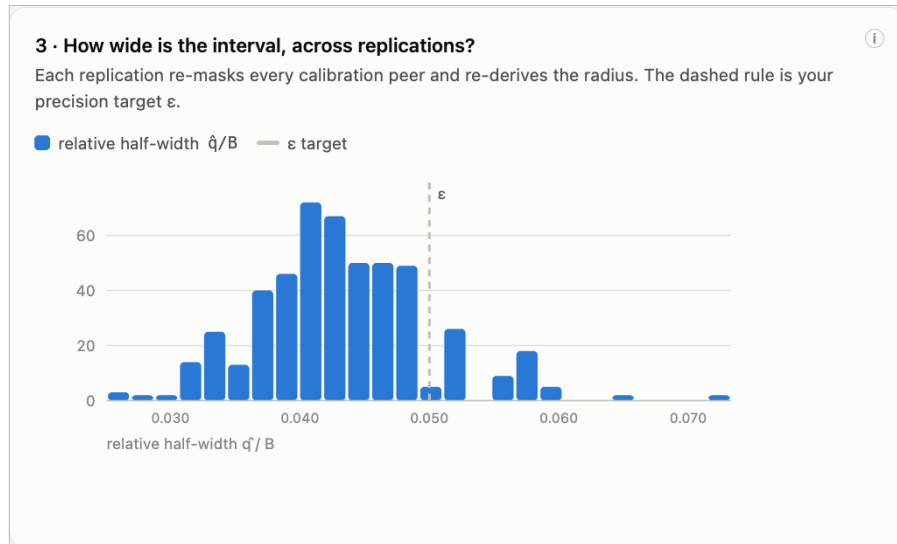
On a month that came in short, the days that never arrived are drawn as flat grey stubs, and clicking one does nothing — there is no reading to switch on. That is not a limitation of the tool. It is the situation a planning threshold exists for: the days are gone, and the question is whether what remains is enough.

**The recommendation does not change with the state of the month.** It is the same number for a complete April, a half-missing April and an April that never reported, because it says how many valid days that station needs in that calendar month *next* time. What the state changes is how much the tool can show you about the month in hand.

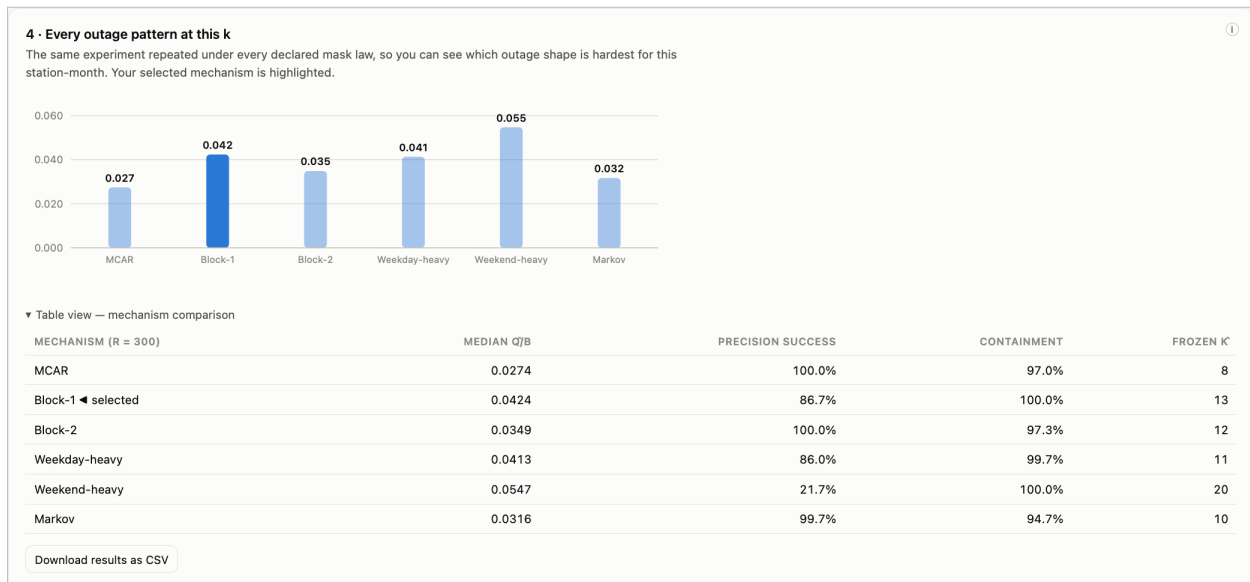
## 9 The five tabs

### 9.1 Run an experiment

The hands-on view, and the one Section 3 walked through. Below the calendar and the prediction interval sit two more panels.



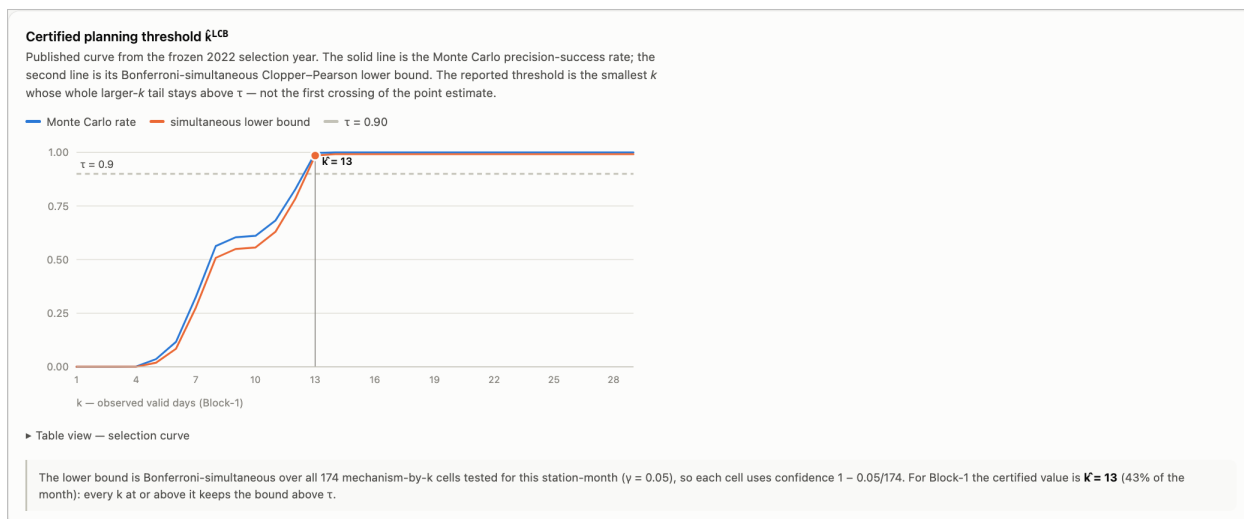
**How wide is the interval, across replications?** The width is not one fixed number — a different set of missing days gives a slightly different margin. This shows the whole spread across the simulated outages, with your  $\epsilon$  target drawn as a rule. The share of the distribution left of that rule is the precision-success rate quoted on the badge.



**Every outage pattern at this k.** The same experiment repeated under all six patterns, so you can see which outage shape is hardest for this particular station. No one pattern is always the worst; it depends on the station's weekly rhythm and on how many days are left.

A **Download results as CSV** button at the foot of the tab saves the per-pattern table for the current station-month.

## 9.2 Planning threshold — where the recommended number comes from

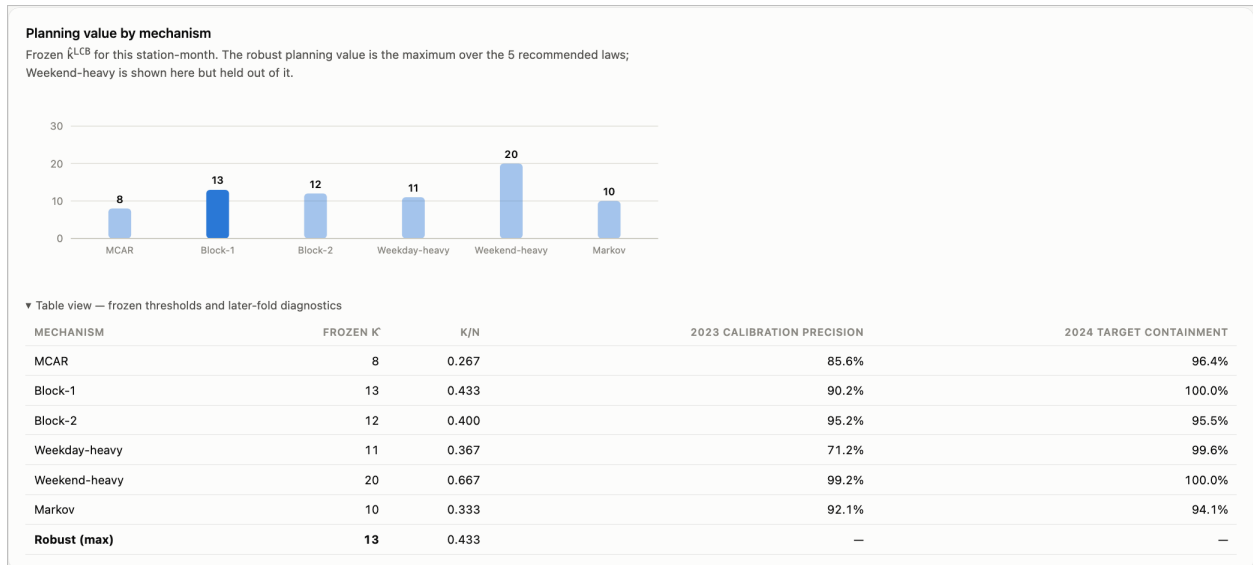


This is the evidence behind the headline number. The curve rises as more days survive: it is the share of simulated outages for which the interval came out tight enough. The dashed rule is the target,  $\tau = 0.90$ .

**The recommendation is not the first point that crosses the target.** It is the first point beyond which the curve *stays* above it, all the way up. That distinction is deliberate: a single lucky day count cannot set the number.

Two lines are drawn. The upper one is the raw simulated rate; the lower one is a conservative bound on it, which is what the threshold is actually read from. The gap between them is the price of being sure rather than merely hopeful.

Open **Table view** beneath the curve and the same two lines appear as numbers, one row per day count. *Successes* / *R* is how many of the simulated outages came out tight enough, out of how many were run; *Rate* is that as a fraction, which is the upper line; and *Simultaneous LCB* is the conservative bound, which is the lower one. LCB is short for lower confidence bound.



The second panel puts all six patterns side by side for this station-month. For 153-0189 in April: MCAR needs 8 days, Markov 10, weekday-heavy 11, Block-2 12, Block-1 13 — and weekend-heavy 20. The **Robust (max)** row at the foot of the table reports **13**, the largest of the five ordinary patterns. Section 11 explains why the 20 does not set it.

That table carries two further columns, both of them *descriptions of what happened* rather than promises. *2023 calibration precision* is how often the interval came out tight enough in the year the margin was calibrated, and *2024 target containment* is how often it actually contained the true average in the year that followed. They are worth looking at, and they are *not* the guarantee: containment is measured on the months this station happened to have, not asserted for the next one.

9.3 Reference pool — who the comparable stations are

Resolved reference cohorts

These are the actual stations whose records become conforal scores. They are chosen by the predeclared cluster → road-family → topology hierarchy using only completeness indicators, never traffic outcomes. The target station is always left out.

| PREDECLARED SCOPE             | 2022 PEERS | 2023 PEERS | STATUS          |
|-------------------------------|------------|------------|-----------------|
| cluster <span>selected</span> | 41         | 43         | preferred (≥19) |
| road family                   | 85         | 90         | preferred (≥19) |
| topology                      | 187        | 183        | preferred (≥19) |

The hierarchy takes the **narrowest** scope with at least 19 distinct stations in both roles; failing that, the scope maximising the smaller cohort provided it reaches 9 — the finite-radius floor for 90% split conforal. Cluster id m04:two\_way:arterial:n247.

Threshold-selection cohort — 2022

Conforal calibration cohort — 2023

| STATION  | FACILITY | FC | MADT 2022 | BASELINE B | STATION  | FACILITY | FC | MADT 2023 | BASELINE B |
|----------|----------|----|-----------|------------|----------|----------|----|-----------|------------|
| 015-0178 | two way  | 3  | 12,572    | 10,797     | 013-0036 | two way  | 4  | 7,937     | 8,426      |
| 021-0116 | two way  | 4  | 4,129     | 4,212      | 015-0178 | two way  | 3  | 12,803    | 10,797     |
| 021-0158 | two way  | 3  | 25,355    | 29,433     | 021-0116 | two way  | 4  | 3,944     | 4,212      |
| 021-0258 | two way  | 3  | 21,639    | 25,179     | 021-0267 | two way  | 4  | 15,210    | 15,793     |
| 021-0541 | two way  | 4  | 17,392    | 18,854     | 021-0541 | two way  | 4  | 16,711    | 18,854     |
| 021-0587 | two way  | 4  | 8,046     | 8,708      | 021-0587 | two way  | 4  | 6,851     | 8,708      |
| 023-0152 | two way  | 4  | 2,960     | 3,496      | 023-0152 | two way  | 4  | 2,898     | 3,496      |
| 051-0264 | two way  | 3  | 26,962    | 22,477     | 051-0264 | two way  | 3  | 25,944    | 22,477     |
| 051-0649 | two way  | 4  | 3,585     | 3,759      | 051-0649 | two way  | 4  | 3,433     | 3,759      |
| 059-0118 | two way  | 4  | 22,738    | 24,344     | 059-0118 | two way  | 4  | 22,534    | 24,344     |
| 059-0367 | two way  | 4  | 12,279    | 12,978     | 059-0367 | two way  | 4  | 12,465    | 12,978     |
| 063-1172 | two way  | 4  | 20,988    | 20,125     | 063-1032 | two way  | 4  | 19,539    | 19,140     |
| 067-0000 | arterial | 0  | 11,000    | 15,000     | 067-0000 | arterial | 0  | 20,000    | 21,000     |

Every margin of error in this tool is learned from how far the method missed at *comparable* stations. This tab names them: the stations used to select the threshold, the stations used to calibrate the interval, and which grouping they were drawn from.

The three scopes at the top

Comparable can be drawn tightly or loosely, so the tool tries three predeclared groupings and reports all of them:

| Scope       | Who is in it                                                                                                                      |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| cluster     | The closest grouping: stations of the same kind on the same family of road whose traffic shapes were grouped together in advance. |
| road family | Wider: the same kind of station on the same family of road, without the finer grouping.                                           |
| topology    | Widest: any station of the same kind.                                                                                             |

The two number columns are how many comparable stations that scope actually supplies, in 2022 and in 2023. The rule then picks the **narrowest** scope with at least 19 in both years; if none reaches 19, it takes the one that leaves the larger cohort, provided it reaches 9. Nine is a hard floor — below it no usable interval exists at all — and 19 is preferred because the width steadies. The chosen row is badged *selected*, and the status column says which of the two bars it cleared.

Closing that paragraph is a **cluster id** such as m04:two\_way:arterial:n247. It is an internal label for the exact group, readable left to right as the month, the kind of station, the road family,

and an identifier for the grouping itself. You never need it to use the tool; it is there so a result can be traced back to the grouping that produced it.

## What the tab guarantees

- **The station is never in its own comparison group.**
- Groups are formed in advance from years older than anything being reported, so a target outcome can never influence its own peer set.
- The route-aligned companion list at the foot of the tab is an engineering diagnostic only. It never decides eligibility, ordering, or stopping, and it is not the comparison group. Its *tier* column says why each station is listed: `lrs_corridor` for the same physical corridor, `posted_route` for the same route number, and `cluster_supplement` for one drawn from the same grouping to fill the list out.

## 9.4 Verification — the page checking itself

**In-browser verification**  
 This page recomputes the manuscript's arithmetic in JavaScript, so it checks itself against the frozen published results shipped in the bundle. Every check below runs live in your browser right now.

9 of 9 checks pass

|                                                                                                                                   |      |
|-----------------------------------------------------------------------------------------------------------------------------------|------|
| Persistent-tail rule reproduces every published $K$<br>16776 / 16776 mechanism-months                                             | PASS |
| Simultaneous Clopper–Pearson bound recomputed in-browser<br>50,328 sampled cells · max $ \Delta  = 5.00\text{e-}7$                | PASS |
| MADT equals the full-calendar mean of the facility-day vector<br>6,967 complete station-months · max $ \Delta  = 5.82\text{e-}11$ | PASS |
| Expanded template has unit mean ( $H = 1$ )<br>1,385 peer templates · max $ \Delta  = 6.66\text{e-}16$                            | PASS |
| Profile-ratio rule on a full mask returns the exact MADT<br>max $ \Delta  = 1.25\text{e-}13$                                      | PASS |
| Conformal radius takes order statistic $r(n+1)(1-\alpha)$<br>$n = 9\ldots 31$                                                     | PASS |
| Finite-radius floor: $n = 8 \rightarrow \infty$ , $n = 9 \rightarrow \text{finite}$<br>$\alpha = 0.10$ , matching Proposition 3   | PASS |
| Every mask kernel returns exactly $k$ observed days<br>6 laws × 3 calendars × 4 values of $k$                                     | PASS |
| Every cohort clears the floor, is distinct-station, and excludes the target<br>2796 station-months checked                        | PASS |

These are exact, deterministic identities — not statistical agreement. If any row fails, the numbers on the other tabs should not be trusted.

This tab re-derives the published arithmetic in your browser and shows it agreeing with the delivered values. These are exact, deterministic identities, not statistical agreement: if any row fails, a number on the page is wrong and should not be used. It is here so the tool can be checked rather than taken on trust — on whatever machine it happens to be opened on.

There are **nine** rows, and the heading above them counts how many passed. *9 of 9 checks pass*, with every row green, is what a healthy copy of the tool shows. Anything else — a red row, or fewer than nine rows listed — is worth reporting before the numbers are used.

## 9.5 How to read this — the method note

### What this tool does, and what it does not claim

**The question.** A calendar month at a continuous count station is only partly observed. How many valid days are enough before a prediction interval for the full-month MADT is both trustworthy and tight enough to use?

**What the dials move, and what stays put.** The three statistical settings —  $\alpha$ ,  $\epsilon$  and the number of replications — drive the *experiment*. Change one and the prediction interval, its width, and the success rates are all worked out again from the underlying data. The **recommended planning value** does not follow them. It is the certified result of the frozen study, computed at  $\alpha = 0.10$ ,  $\epsilon = 0.05$ , a **90% success target** and **1,000 replications**, and it is the figure the report and the delivered spreadsheet quote.

This is not because the number is independent of those settings — it plainly is not. Worked example, at station 153-0189 in January: with the delivered settings, 16 observed days give an interval of about  $\pm 2.99\%$  of the station's normal traffic, which clears the 5% requirement every time. Tighten the requirement to  $\epsilon = 0.02$  and that same 16 days clears it *never* — so the honest recommendation at those settings would be a good deal higher than 16. Re-deriving it properly means re-running the entire selection experiment: every day count, every outage pattern, a thousand replications each, with a simultaneous correction across all of those cells at once. That is far heavier than anything computed live in this page, so the certified value is shipped rather than recalculated.

If you want to reason at other settings, set them and then move the day count, watching the *precision success* figure on the interval panel: it tells you how often a month with that many days would come out tight enough under your settings. Treat it as a sound indication rather than a certified value — it comes from a single fold and carries no correction for the many day counts you tried.

**The estimand is the whole facility.** A two-way station's daily total needs both travel directions; a ramp or one-way station uses its single registered stream. A partial directional total is a different quantity and is not used.

**Two separate targets.**  $\alpha$  controls *coverage* — the chance the interval contains the true MADT.  $\epsilon$  controls *precision* — how wide the interval may be, as a fraction of the station's frozen baseline scale.  $\tau$  is the probability of hitting that width target under the declared mask law, and  $\gamma$  is Monte Carlo familywise error. They are four different numbers and are never interchangeable.

**The chronology is what keeps it honest.** Profiles, templates, baselines, and peer groups come from 2017–2019. The planning value is certified on 2022 alone. Conformal calibration happens later, on 2023, at the already-frozen  $k$ . 2024 is held out. A target outcome can never tune a threshold.

**The guarantee.** Conditional on the frozen selection and an explicit fixed-site exchangeability assumption for the normalized scores, the interval has finite-sample marginal coverage of at least  $1 - \alpha$ . That statement is marginal over the declared mask law and is made one site at a time.

**What is not claimed.** No coverage conditional on one realized outage pattern; no simultaneous guarantee across the network; nothing for an undeclared or adversarial mechanism; and no promise that a narrow interval is therefore a covering one. When you hand-edit the calendar you are asking a *realized-pattern* question, which the theorem deliberately does not answer — the tool reports it as a diagnostic and labels it as such.

**A complete month needs no prediction.** If the month is strictly complete its MADT is reported exactly. The planning value stays prospective: it tells you what to expect from a *future* delivery of that calendar month.

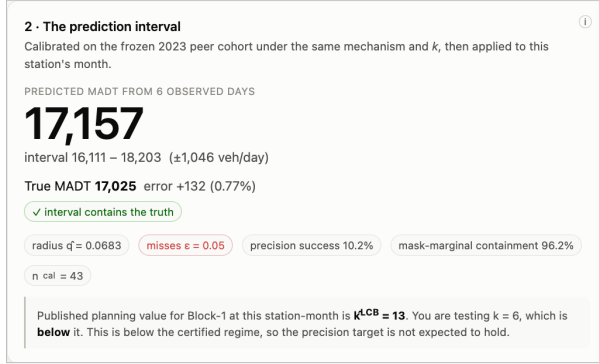
Bundle generated 2026-08-21T19:48:03Z by dashboard/build\_bundle.py from the frozen v2 artifacts behind final\_deliverables/final\_report.tex. Constants:  $\alpha = 0.1$ ,  $\epsilon = 0.05$ ,  $\tau = 0.9$ ,  $\gamma = 0.05$ ,  $R = 1000$ . Peer floor 9, preferred 19.

A condensed statement of what the tool does and does not claim, always available beside the data. Section 12 of this guide is the same material in longer form.

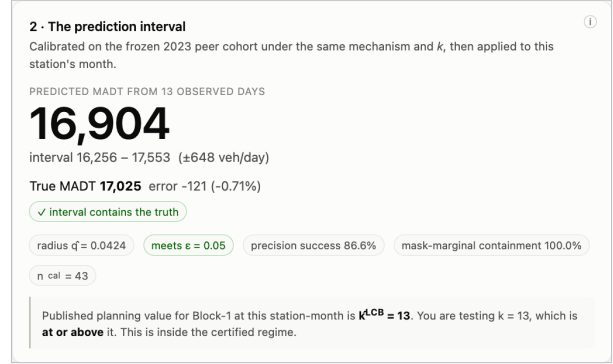
## 10 Seeing why the day count matters

The quickest way to understand the recommendation is to break it. Keep everything else fixed and move the slider.

## 6 days — below the recommendation



## 13 days — at the recommendation



|                                      | 6 days                     | 13 days           |
|--------------------------------------|----------------------------|-------------------|
| Interval half-width                  | $\pm 1,046$ veh/day        | $\pm 648$ veh/day |
| As a share of scale                  | 0.0683                     | 0.0424            |
| Against the $\epsilon = 0.05$ target | <b>misses</b>              | <b>meets</b>      |
| Precision success rate               | 10.2%                      | 86.6%             |
| Verdict                              | below the certified regime | inside it         |

Both intervals happen to contain the true value. That is the point worth understanding: the six-day interval is not *wrong*, it is *too wide to be useful*, and it is too wide far too often — it hits the precision target only about one time in ten. The recommendation is the day count at which that stops being a gamble.

## 11 Weekend-heavy: shown, but not recommended

The recommended planning value is the largest requirement among *five* of the six patterns. Weekend-heavy is deliberately left out of it.

### Why

- **The arithmetic is brutal.** A month contains only eight or nine weekend dates. A pattern that removes those first takes away a whole block of the weekly rhythm at once, leaving the estimator nothing to anchor on, and the required day count jumps.
- **It would dominate almost everything.** Weekend-heavy sets the largest of the six values in roughly **three station-months out of five**, sometimes by as much as **twelve days**. At 153–0189 in April it asks for 20 days where the next hardest pattern asks for 13.
- **It is a rare failure mode.** An outage landing squarely and only on weekends is unusual. Sizing every station-month for it would push the recommendation far beyond what ordinary interruptions require, and a target nobody can meet is not useful guidance.

## What is *not* done

**Nothing is hidden.** Weekend-heavy stays fully selectable, with its own curve and its own certified count. Wherever it would raise the number, the recommended-value tile prints that higher figure beside the recommendation — at 153–0189 in April it reads “20 *incl. weekend-heavy*”. If a station genuinely faces weekend risk, such as a recurring weekend closure or planned weekend maintenance, select weekend-heavy and read its number directly.

## 12 What the tool does not claim

- **It does not produce an interval for a month you already hold.** The recommendation is prospective guidance for a future month of the same name.
- **A recommendation is tied to a stated outage pattern.** It is not a guarantee for an arbitrary pattern of missing days. Hand-editing the calendar is labelled a realized-pattern diagnostic for exactly this reason.
- **The guarantee is per station.** It is invoked one station at a time and is not simultaneous across the network.
- **A complete month is reported exactly.** No prediction replaces it.
- **Precision and coverage are separate.** A narrow interval need not contain the truth, and a covering interval need not be narrow.
- **Comparisons across stations are descriptive.** They do not establish that road class or traffic volume *causes* a requirement.

## 13 If something looks wrong

| Symptom                                              | What it means, and what to do                                                                                                                                |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The recommended value is blank                       | The station-month was not certified at the simulation resolution used. It is a status, not a zero — treat it as “no recommendation”, not “zero days”.        |
| The page is slow on first open                       | It carries all its own data. Give it a few seconds; after the first load it is immediate.                                                                    |
| You see raw markup instead of the dashboard          | The <code>.html</code> file opened in a text editor. Right-click → <b>Open With</b> → your browser.                                                          |
| The numbers moved after you changed a setting        | They should: $\alpha$ , $\epsilon$ , $R$ and the pattern all change the answer. Press <b>Reset to defaults</b> to return to the delivered figures.           |
| A number disagrees with the delivered planning table | The table and this tool use the same recommendation rule and agree on all 2,796 station-months. If they differ, a setting has been changed from its default. |
| A row on the Verification tab fails                  | Stop and report it. Those are exact identities; a failure means a number on the page is wrong.                                                               |

| Symptom                                      | What it means, and what to do                                                                                                                                              |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The page opens but the station list is empty | The file did not arrive whole. It is about 25 MB, and a download that stops early leaves a page that looks right and has no data in it. Fetch it again and check the size. |

## 14 Glossary

| Term                                 | In plain language                                                                                                                                                                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MADT</b>                          | Monthly Average Daily Traffic — the average of the station’s daily totals across a calendar month. The quantity everything here is about.                                                                                                           |
| <b><math>k</math></b>                | The number of valid days a month is left with after losses.                                                                                                                                                                                         |
| <b>Planning value</b>                | The recommended $k$ : the fewest valid days at which the interval is reliably tight enough. Written $\hat{k}^{\text{LCB}}$ on screen.                                                                                                               |
| <b>Prediction interval</b>           | A range around the estimate, built so that it contains the true monthly average at least $1 - \alpha$ of the time.                                                                                                                                  |
| <b><math>\alpha</math></b>           | Miscoverage. How often the interval is allowed to miss. Ships at 0.10.                                                                                                                                                                              |
| <b><math>\epsilon</math></b>         | Relative half-width. How wide the interval may be, as a share of the station’s normal scale. Ships at 0.05.                                                                                                                                         |
| <b><math>\tau</math></b>             | The precision-success target: how often the interval must come out tight enough before a day count is certified. Fixed at 0.90.                                                                                                                     |
| <b>Baseline scale <math>B</math></b> | The station’s normal traffic for that month, fixed from 2017–2019. Widths and errors are quoted against it.                                                                                                                                         |
| <b>Cohort</b>                        | The set of comparable stations the margin of error is learned from.                                                                                                                                                                                 |
| <b>Scope</b>                         | How widely <i>comparable</i> is drawn: the close <i>cluster</i> , the wider <i>road family</i> , or the widest <i>topology</i> . The narrowest one that supplies enough stations is the one used.                                                   |
| <b>Preferred / finite only</b>       | Whether the chosen scope supplied at least 19 comparable stations (preferred, for a steady width) or merely cleared the hard floor of 9 (finite only, usable but read with more caution).                                                           |
| <b>Cluster id</b>                    | An internal label for the exact grouping a result came from, such as <code>m04:two_way:arterial:n247</code> . Useful for tracing a number, never needed to read one.                                                                                |
| <b>Simultaneous bound</b>            | The lower of the two lines on the certification curve. It is lowered on purpose so that being wrong at <i>any</i> of the day counts and patterns tested is unlikely, not merely at one of them. On screen this is called the Bonferroni correction. |
| <b>Mechanism</b>                     | The assumed shape of the outage — which days went missing, not how many.                                                                                                                                                                            |

| Term             | In plain language                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conformal</b> | The family of methods used here. Its intervals are built by ranking how far the method missed on past comparable cases, rather than by assuming a model for traffic. |